London Ambulance Service NHS
NHS Trust

Diabetes

Component 5

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London Ambulance Service NHS Trust

Glycaemic Emergencies (Diabetes) - Component 5

- What is diabetes
- Who is affected
- Causes
- Diabetic Emergencies
- Assessment and Management

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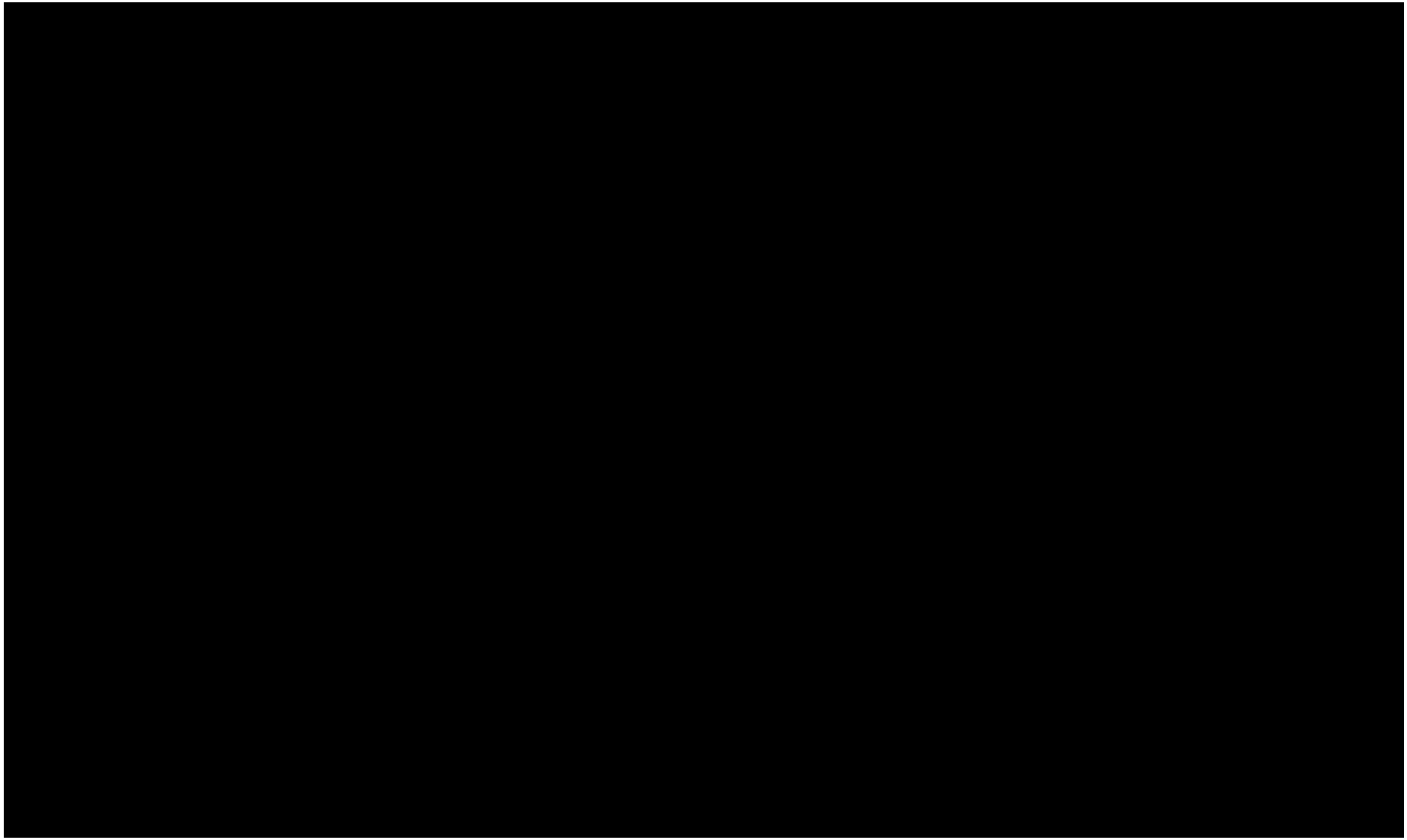
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Diabetes

A common metabolic disorder identified by chronic high blood sugar levels.

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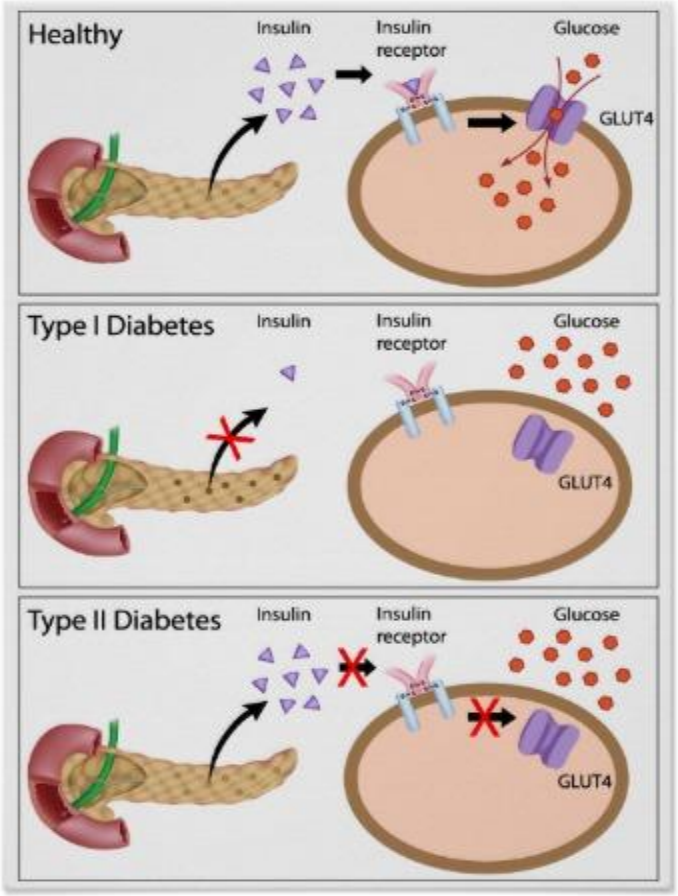
Underlying causes of diabetes mellitus generally fall into 4 categories

- Genetic
- Obesity
- Age
- Ethnicity

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What is happening in Diabetes?



- **In Type 1 Diabetes** Insulin is not being produced at all

- **In Type 2 Diabetes** Insulin is released but the receptor sites have mutated and Insulin is unable to enter the cells to allow sufficient Glucose absorption

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How is Diabetes Controlled?

Type 1



Type 2







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Blood Glucose Levels

Blood glucose level is the amount of glucose (sugar) in the blood. This is expressed as millimoles per litre (mmol/L)



<u>Normal values:</u>

- Non-diabetic 3.0 to 5.6mmol/L
- Diabetics 4 to 7mmol/L before meals.
- Less than 10mmol/L one and a half hours after meals.
- Around 8mmol/L at bedtime.

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Hypoglycaemia

Common causes of hypoglycaemia:

- Overdose of Insulin or oral hypoglycaemic agents
- Hepatic or Renal dysfunction
- Other illnesses, including infection
- Gastrointestinal disturbances
- Inadequate food intake
- Increased physical activity
- Alcohol intake
- Pregnancy and breastfeeding
- The elderly

Hypoglycaemia is a BM <4mmol. Treatment must take place if the BM is suspected or confirmed to be less than 4.

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Hypoglycaemia signs and symptoms

Hypoglycaemia may occur in Diabetics or other patients who are unwell and can display one or more of the following signs:

- **Autonomic;** sweating, palpitations, shaking, hunger
- **Neurological;** confusion, drowsiness, odd behaviour, speech difficulty, un-coordination, aggression, fitting, unconsciousness
- **General malaise;** headache, nausea

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Management of Hypoglycaemia

What does JRCALC PLUS SAY?



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Hyperglycaemia

is a term used to describe high blood sugar levels

Causes of Hyperglycaemia :

- Undiagnosed diabetes
- Illness
- Infection
- Myocardial infarction
- Medication including steroids
- Failure to administer diabetic medications

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Hyperglycaemia signs and symptoms

- Polyuria
- Polydipsia
- Polyphagia
- Ketone breath
- Kussmaul's respiration
- Lethargy and confusion
- Dehydration and signs of shock
- Hyperventilation
- Rapid Weight loss
- Unconsciousness

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Management of Hyperglycaemia

- Correct ABC problems
- Measure and record BM
- Record ECG
- Administer supplemental Oxygen if hypoxaemic
- Pre-alert

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Measuring blood glucose

- Clean patient's finger using soap and water or alcohol swab wipe for injection
- **DO NOT USE CHLOROPREP**
- Once finger is dry obtain a blood sample using capillary lancet
- Wipe away the first drop of blood
- Obtain a fresh test strip from the pot and insert black wide end into meter



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Measuring blood glucose continued

- Wait for blood droplet screen to appear after a short self test and apply a drop of blood into the sample end 
- The result will be displayed within 5 seconds in mmol/L
- Press eject button whilst holding meter over clinical waste bin to ensure test strip is disposed of safely

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Summary – Hypo V Hyperglycaemia

	Hypoglycaemia	Hyperglycaemia
Onset	Rapid (within minutes)	Gradual (hours to days)
Skin	Pale & sweaty	Dry
Respiration	Normal or shallow	Deep & sighing
Pulse	Rapid & full	Rapid & weak
Behaviour	Irritable, confused, uncoordinated, aggressive / anti-social	Restless, drowsy or lethargic
Others	Hunger Headache Dizziness Seizures in late stages	Sweet fruity breath Abdominal pain Nausea Excessive urine Excessive thirst Hypotension

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Questions?



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